

DENGUE OUTBREAK RISK PREDICTION

Project Proposal

Group 4

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The Problem

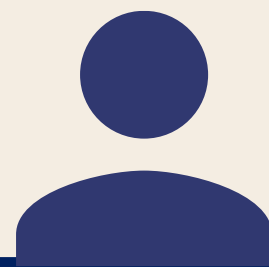
Singapore's 2020 dengue outbreak saw over **35,000 cases**, yet the tools used to respond were **largely reactive**.

There is **no system** to predict which neighborhoods were at risk before clusters formed, leaving National Environment Agency (NEA) and Communicable Diseases Agency (CDA) **struggling to catch up** rather than getting ahead of the outbreak

Proposed Solution

Dengue Control is a *Two Speed Problem* | *Tow Cost Functions* | *Two Cadences* | *One underlying Model Serving Both*

Strategic Loop



NEA Operations Team

Primary User

Where should fogging be deployed next week ?

When

Weekly – Every Monday Morning

Receives

Model-predicted risk score per subzone

Action

Pre-deploy fogging teams to High-risk zones

Tactical Loop



CDA Officer at MOH

Real Time New User

Who needs same day outreach right now ?

When

Event Driven – Per confirmed case

Receives

ML vulnerability score and 200m radius features

Action

Triage: Home Visit, Advisory, Testing van dispatch

The Dataset

ANCHOR

SGCharts Dengue Cluster Snapshots

May 2013 → Nov 2020 | 250+ bi-weekly snapshots | Community CSV

- Lat/long cluster boundaries and case counts
- 7 years of real spatial cluster data
- Covers 2020 record outbreak (35k+ cases) - concept drift event

AUXILIARY | Weather

MSS Weather Station Records

Met Service Singapore | Station-level Daily | 2013 - Present

- Rainfall, temperature, relative humidity per station
- Replaces NEA realtime API; extends usable training window
- Drives lag features: 1w / 2w / 4w rainfall, 4w temp mean

AUXILIARY | Demographics & Geometry

SingStat Census and URA Master Plan

Census 2020 | URA Master Plan 2019 GeoJSON

- Subzone-level population, age, comorbidity proxies
- Vulnerability index (% elderly + chronic) for tactical layer
- URA polygons → spatial join target for label derivation

LABEL

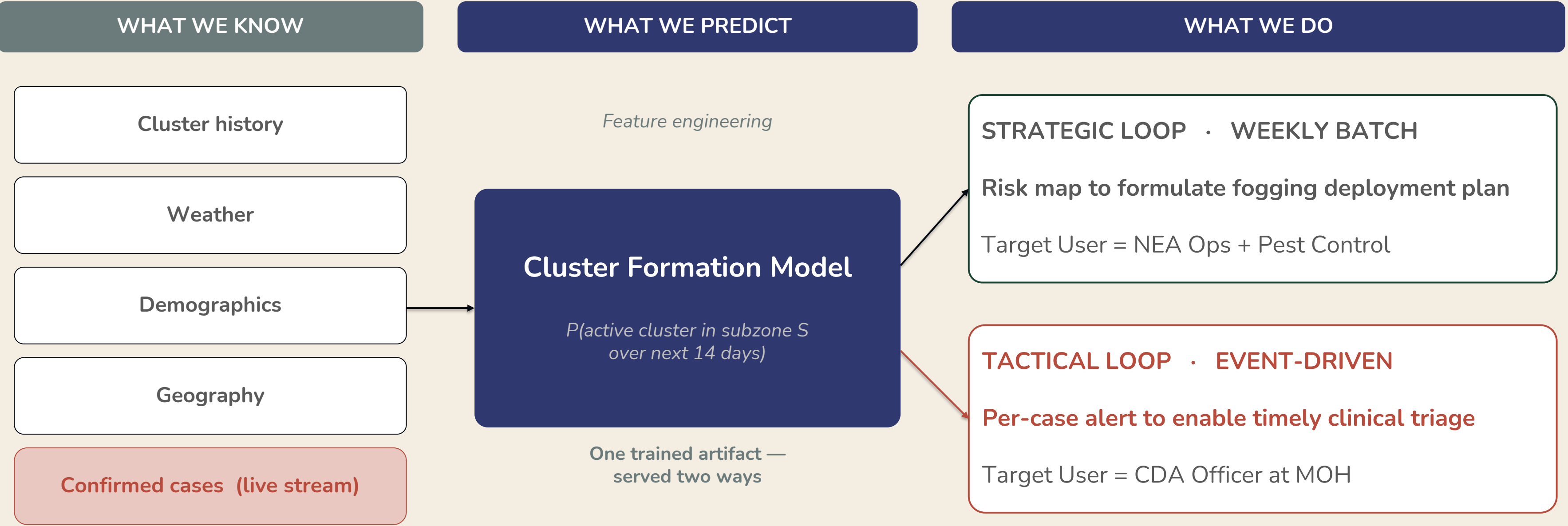
Spatial Join → Binary Label

Derived from Anchor | Fully Transparent Process

- URA subzones intersected with SGCharts cluster polygons
- Label = 1 if active cluster in subzone in next 14 days
- No synthetic data; every label traces to real cluster

One model, two inference loops

Shared predictive logic; different feature freshness, different decision thresholds, different downstream actions.



Same predictive logic. Different freshness contracts. Different thresholds. Different actions.

Design Choices

WHAT WE DECIDED FROM BEGINNING

✓ Training window: 2013–2020

Use full SGCharts window. 2020 serotype shift (DENV-3, 35k+ cases) demonstrates concept drift in the report.

✓ Subzone granularity: 330

Fine-grained for operational fogging. Planning areas (55) lose deployment precision.

✓ Weather source: MSS, not NEA API

MSS station data extends usable window back to 2013. NEA API has Dec 2016 floor + 19-hr ingest.

DESIGN CHOICES AND RATIONALE

Two-layer by design: batch + real-time

Different operational loops, different cost functions. Same trained model, two inference paths.

XGBoost / LightGBM, not deep learning

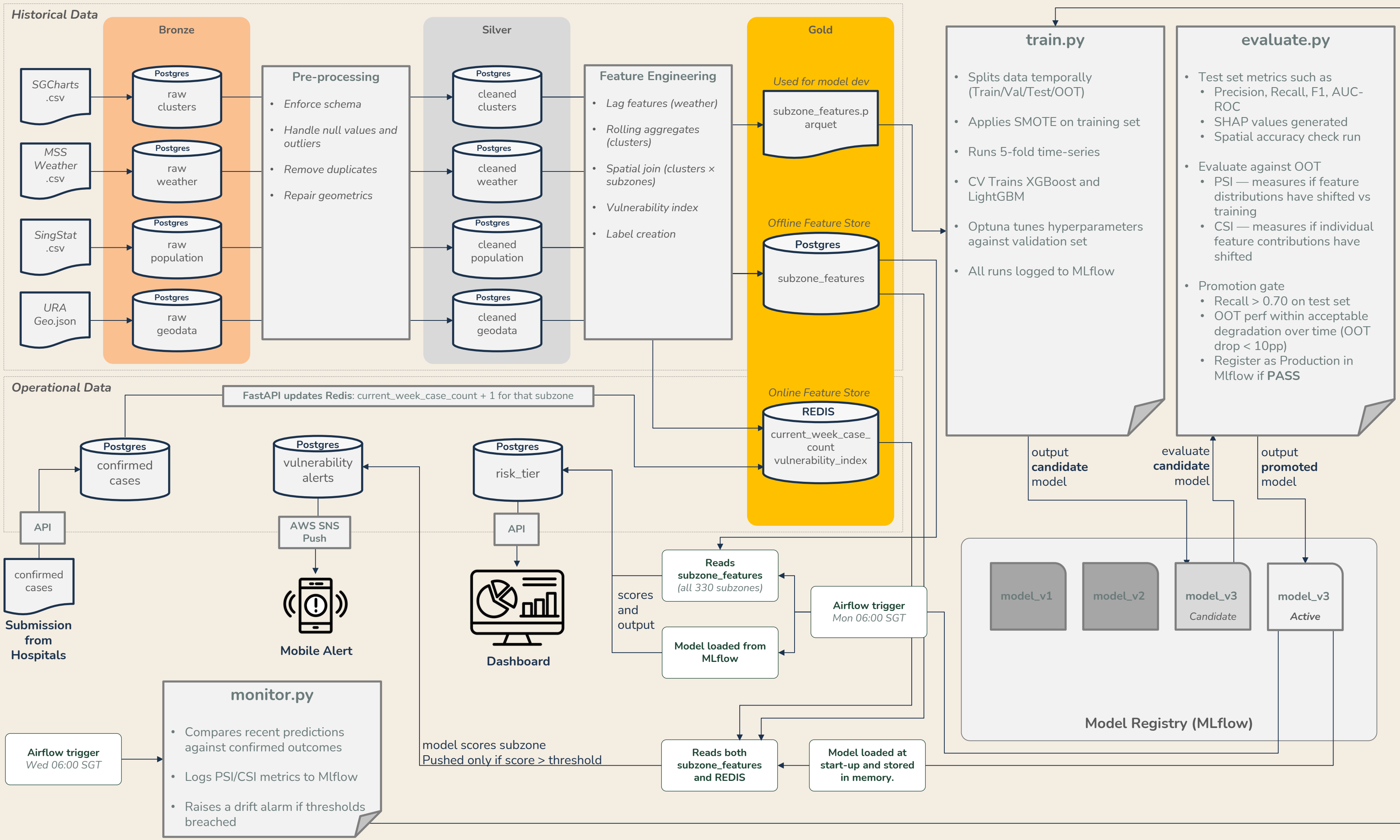
Tabular spatial+temporal data. Interpretable via SHAP. Trains in minutes locally.

Time-series sliding-window CV

Standard k-fold leaks future cluster data, temporal CV is the only valid choice.

Recall ≥ 0.70 on High-risk tier

Asymmetric cost: false negative = missed outbreak. We optimise for the more expensive error.



Thank you!

For guidance ;)